

Índice del tema *ii*

▶ *ii.* Microhidrodinámica

- ▶ Fundamentos y abordaje académico
- ▶ Navier-Stokes, regímenes de flujo
- ▶ Flujos 1D: perfiles de velocidad
- ▶ Modelo generalizado
- ▶ Relaciones flujo-fuerza
- ▶ Redes de microcanales
- ▶ Co-flujos (juntura Y, juntura X)
- ▶ Minimización de la resistencia (Murray)
- ▶ Flujos 2D y 3D: simulación

ii. ¿Para qué estudiar los fundamentos?

**fundamentos
y modelado**



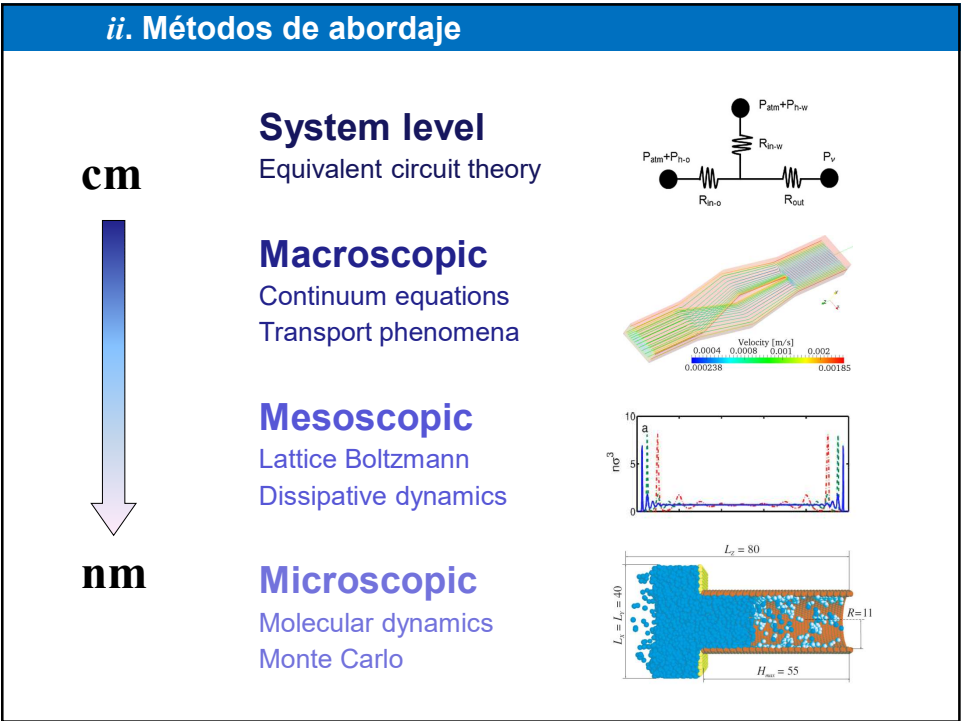
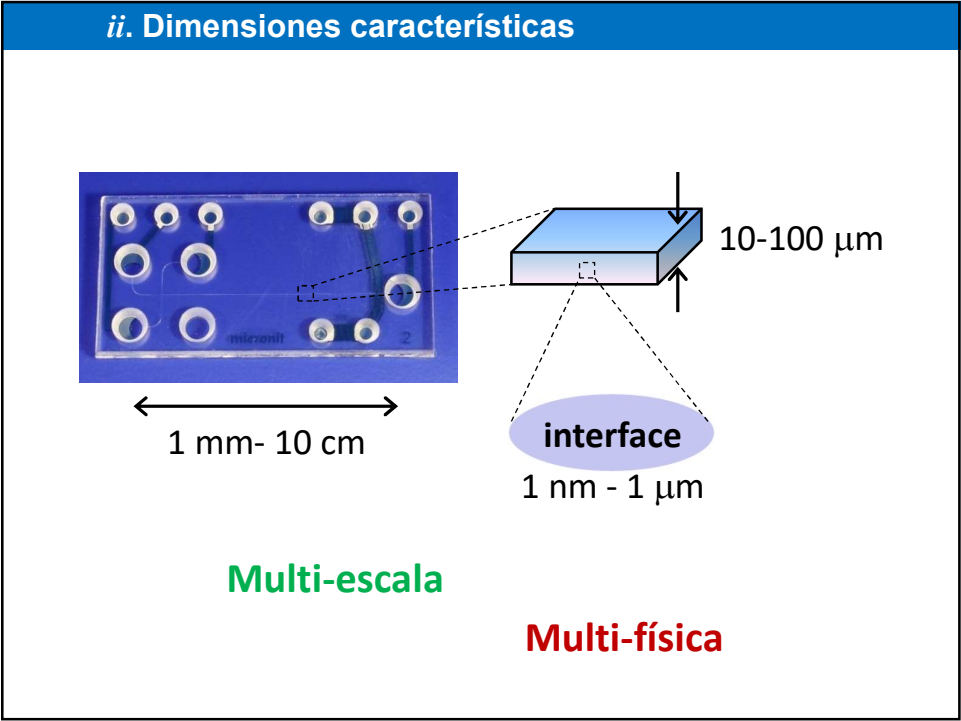
**simulación,
diseño y
fabricación**



**operación
de los chips**



nuevos conocimientos



ii. Breve historia académica

Happel and Brenner (1965) **Low Re Number Hydrodynamics**

Batchelor (1977) **Microhydrodynamics**

Probstein (1989) **Physicochemical Hydrodynamics**

Karniadakis and Beskok (2002) **Microflows**

Tabeling (2005) **Introduction to Microfluidics**

Bruus (2008) **Theoretical Microfluidics**

Lab-on-a-Chip

(RCS, 2000)

Microfluidics and Nanofluidics

(Springer, 2004)

Biomicrofluidics

(AIP, 2007)

ii. Reynolds, Navier y Stokes

$$\nabla \cdot \mathbf{u} = 0$$

Newtoniano
incompresible
isotérmico

Navier, 1785-1836
Stokes, 1819-1903

$$\rho \left(\frac{d\mathbf{u}}{dt} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla P + \mu \nabla^2 \mathbf{u}$$

Reynolds, 1842-1912



$$\text{Re} = \frac{\rho u h}{\mu}$$

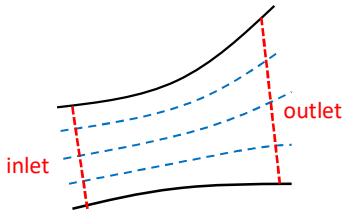
ii. Conservación de materia

$$\nabla \cdot \mathbf{u} = 0$$

Newtoniano
incompresible
isotérmico

$$\frac{du_x}{dx} + \frac{du_y}{dy} + \frac{du_z}{dz} = 0$$

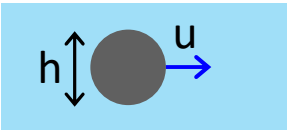
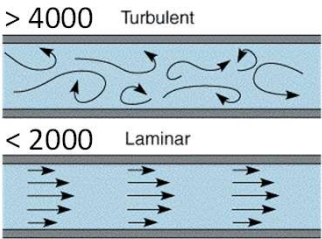
$$\iiint \nabla \cdot \mathbf{u} dV = \iint \mathbf{u} \cdot \mathbf{n} dS$$



$$\iint_{A_{in}} \mathbf{u} \cdot \mathbf{n} dS + \iint_{A_{out}} \mathbf{u} \cdot \mathbf{n} dS = 0$$

ii. Reynolds, Navier y Stokes

$$Re = \frac{\rho u h}{\mu}$$



- > 90 separated
- > 5 unseparated
- < 1 creeping

Re < 1

- Regimen de Stokes
- Sin aceleración, reversible
- Alta viscosidad, baja velocidad
- → o altamente confinado

ii. Regímenes de flujo



$$Re \sim 10^3$$

- Inviscido
- Energía mecánica
- En tuberías: Bernulli, 1739



$$Re \sim 10^{-5}$$

- Velocidad constante
- Microorganismos
- Microcanales

ii. Regímenes de flujo



$$Re \sim 10^{-2}$$

Sistema
macroscópico
equivalente



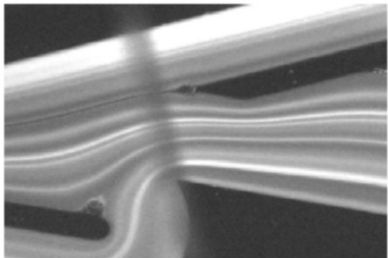
$$\tau_s = \frac{\rho h^2}{\mu} \sim \text{ms}$$

ii. Régimen de Stokes (sin aceleración)

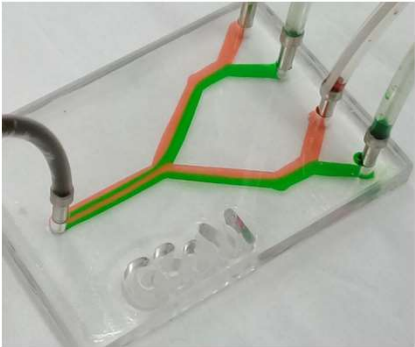
$$\nabla \cdot \mathbf{u} = 0$$

Newtoniano
incompresible
isotérmico

$$0 = -\nabla P + \mu \nabla^2 \mathbf{u}$$

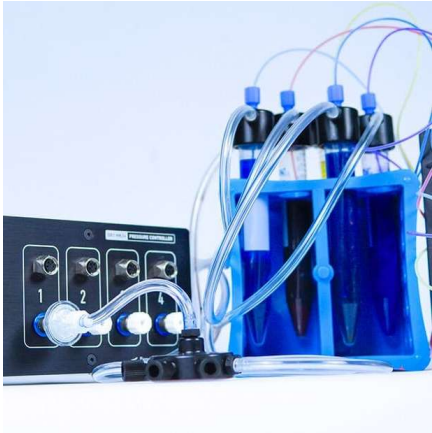


Tabeling lab, Paris



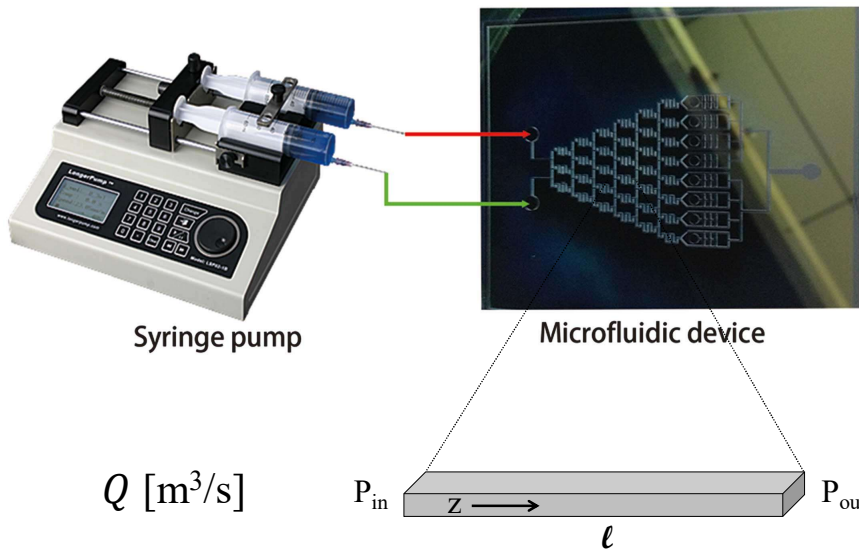
ii. Flujos de presión controlada

ΔP [Pa]



elveflow.com

ii. Flujos de caudal controlada



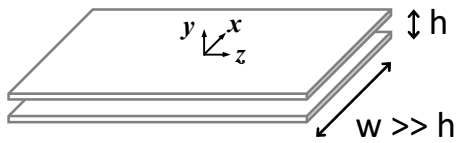
ii. Flujo entre placas planas (slits)

$$\nabla P = \mu \nabla^2 \mathbf{u}$$

$$Re \ll 1$$

Hele-Shaw, 1854-1941

$$\frac{\partial P}{\partial z} = \mu \frac{\partial^2 u_z}{\partial y^2}$$

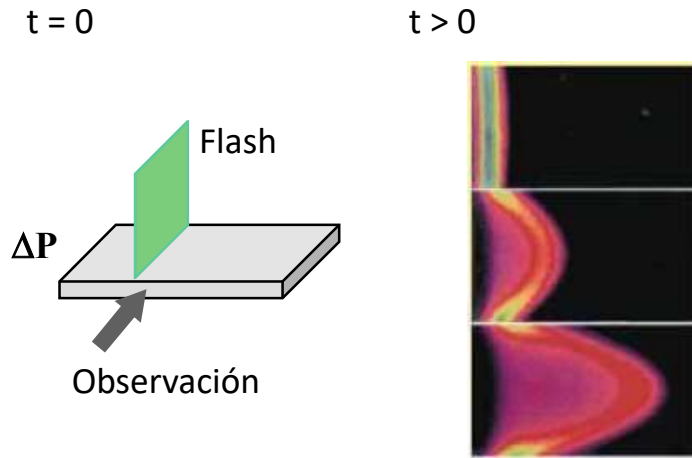


$$y = \pm h/2, \quad u_z = 0$$

$$y = 0, \quad \partial u_z / \partial y = 0$$

$$\Rightarrow u_z(y) = \frac{h^2 \Delta P}{8\mu \ell} \left(1 - \frac{4y^2}{h^2} \right)$$

ii. Perfiles de velocidad



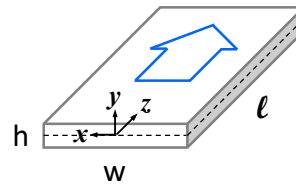
Whitesides and Stroock,
Phys. Today **54** (2002) 42

ii. Caudal volumétrico

$$Q = w \int_{-h/2}^{h/2} u_z(y) dy$$

$$Q = wh \langle u_z \rangle$$

$$\langle u_z \rangle = \frac{1}{h} \int_{-h/2}^{h/2} u_z(y) dy$$



Relación flujo-fuerza lineal:

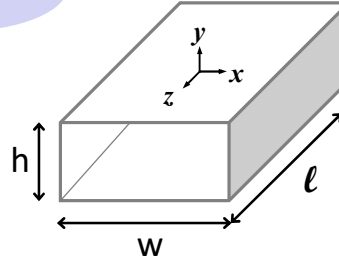
$$Q = \frac{wh^3}{12\mu\ell} \Delta P$$

ii. Flujo en canales rectangulares

$$\nabla P = \mu \nabla^2 \mathbf{u}$$

$$Re \ll 1$$

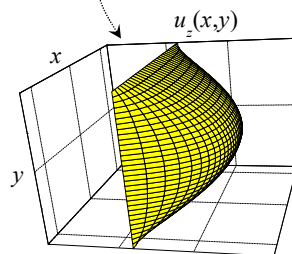
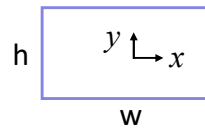
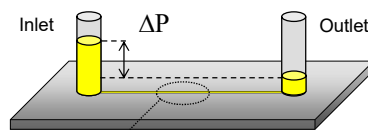
$$\frac{\partial P}{\partial z} = \mu \left(\frac{\partial^2 u_z}{\partial x^2} + \frac{\partial^2 u_z}{\partial y^2} \right)$$



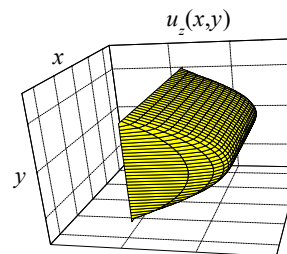
$$\Rightarrow u_z(x, y) = \frac{h^2 \Delta P}{8\mu \ell} G(x, y)$$

$$G(x, y) = 32 \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{\beta_m^3} \left(1 - \frac{\text{ch}(\beta_m x/h)}{\text{ch}(\beta_m w/2h)} \right) \cos(\beta_m y/h)$$

ii. Perfiles de velocidad



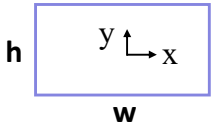
$$h/w = 2/5$$



$$h/w = 1/10$$

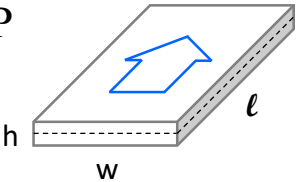
ii. Caudal volumétrico

$$Q = \iint u_z(x,y) dx dy$$



$$= \frac{wh^3 \Delta P}{12\mu \ell} \sum_{m=1}^{\infty} \frac{96}{\beta_m^4} \left[1 - \frac{\tanh(\beta_m w/2h)}{\beta_m w/2h} \right]$$

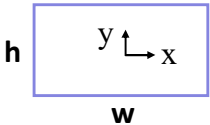
$$Q \approx \frac{wh^3}{12\mu \ell} \Delta P$$



h/w	
0.42	1/1
0.75	2/5
0.99	1/60

ii. Caudal vs presión

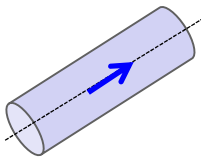
$$Q = \frac{wh^3 f_G}{12\mu \ell} \Delta P$$



$$Q = L \Delta P$$



$$\Delta P = R Q$$



$$R = \frac{8\mu \ell}{\pi r^4}$$

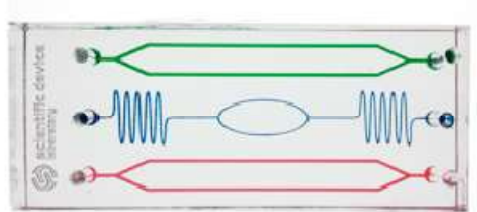
Hagen, 1838
Poiseuille, 1839

ii. Microcanales en serie/paralelo

Caso más simple: **un puerto de entrada y uno de salida**

$$Q = L_{\text{tot}} \Delta P$$

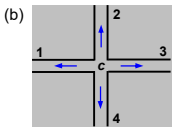
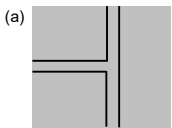
$$\Delta P = R_{\text{tot}} Q$$



$$R_{\text{ser}} = \sum_i R_i$$

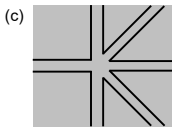
$$R_{\text{par}}^{-1} = \sum_i R_i^{-1}$$

ii. Redes de microcanales



N canales/nodo

→ 1 incógnita

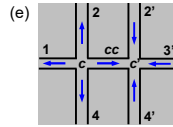
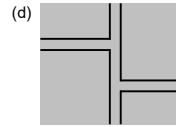


$$\sum_{i=1}^N Q_i = 0$$

$$\sum_{i=1}^N L_i (P_i - P_c) = 0$$

$$P_c = \sum_{i=1}^N L_i P_i / \sum_{i=1}^N L_i$$

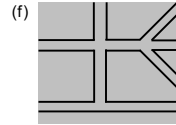
ii. Redes de microcanales



N canales/nodo

M nodos/red

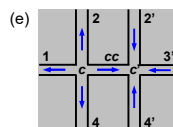
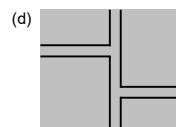
→ **M incógnitas**



$$\sum_{i=1}^N Q_i = 0, \quad j = 1 \text{ a } M$$

$$\begin{bmatrix} \sum_{i=1}^N L_{11,i} & -L_{11,cc} \\ -L_{11,cc} & \sum_{i'=1'}^{N'} L_{11,i'} \end{bmatrix} \begin{bmatrix} P_c \\ P_{c'} \end{bmatrix} = \begin{bmatrix} \sum_{i \neq cc}^N L_{11,i} P_i \\ \sum_{i' \neq cc}^{N'} L_{11,i'} P_{i'} \end{bmatrix}$$

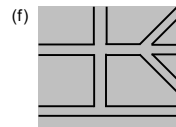
ii. Redes de microcanales



N canales/nodo

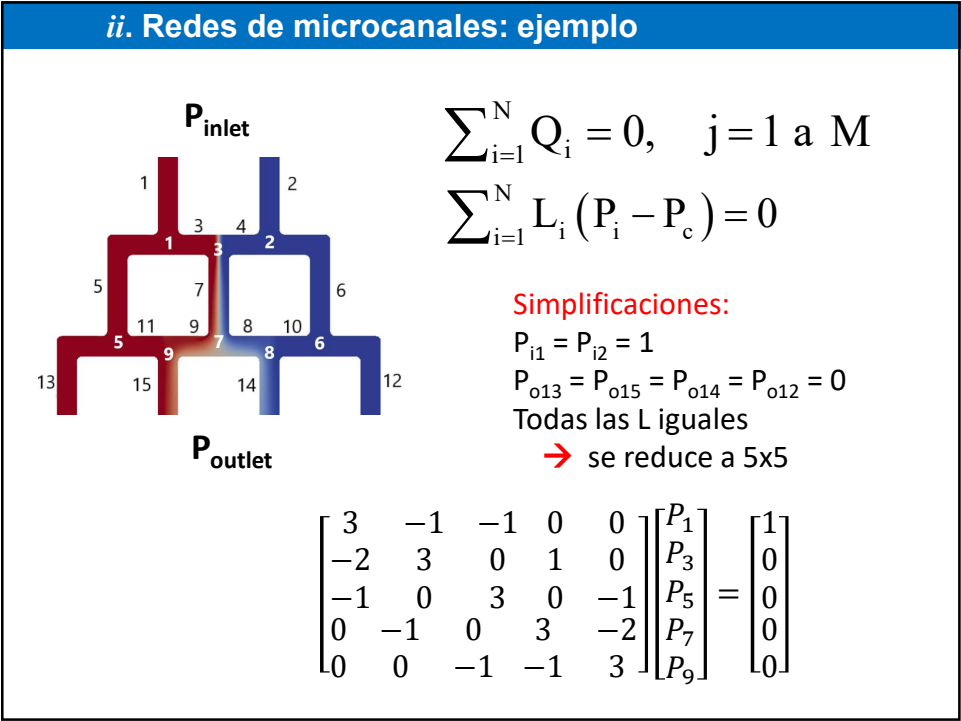
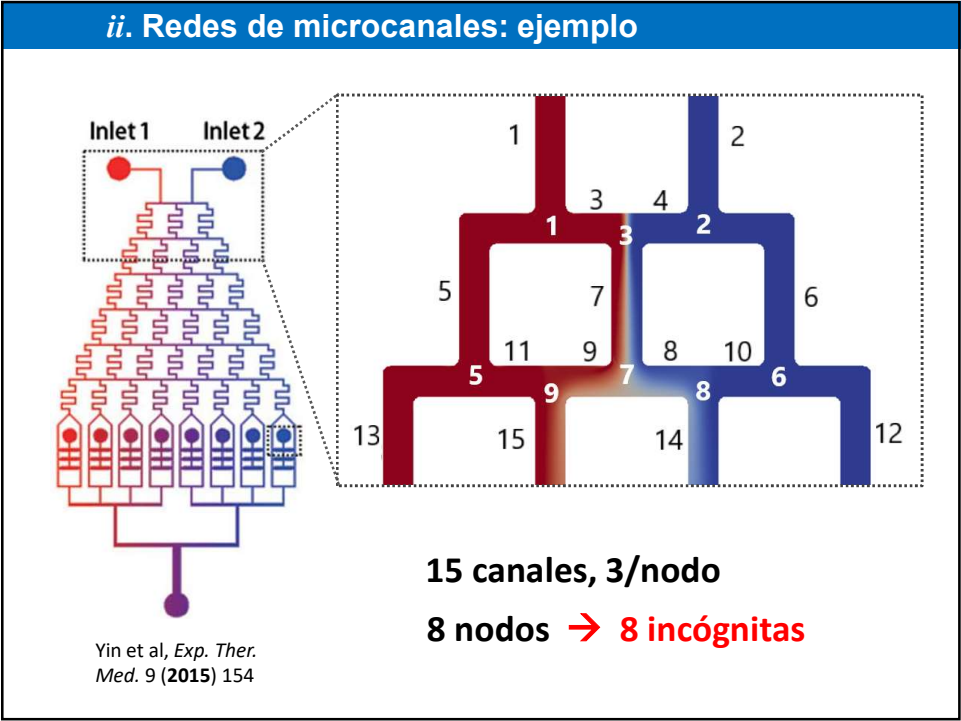
M nodos/red

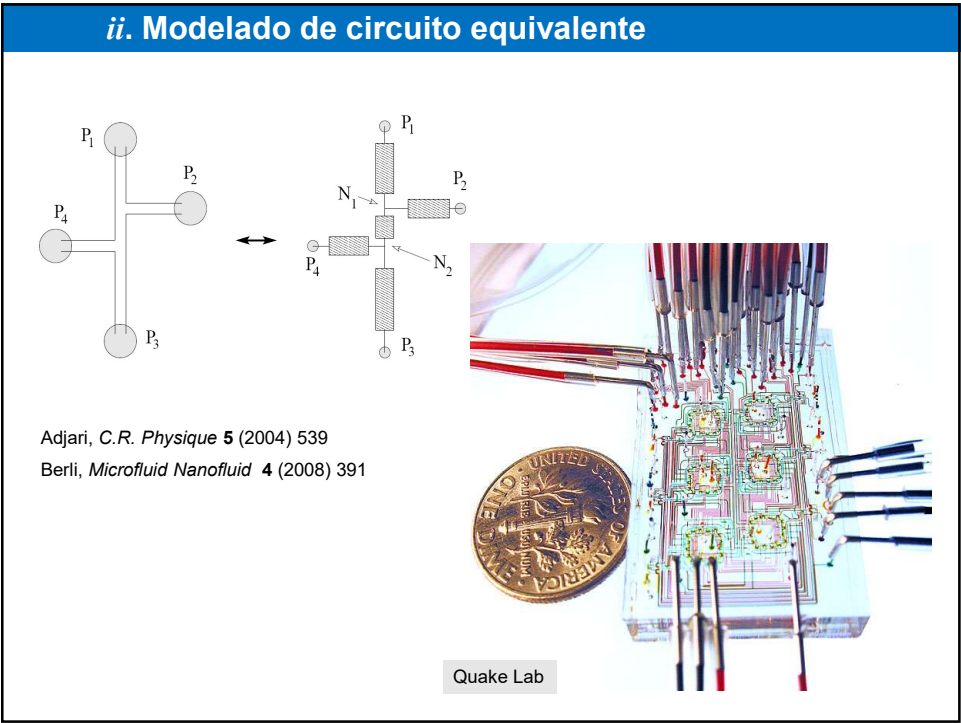
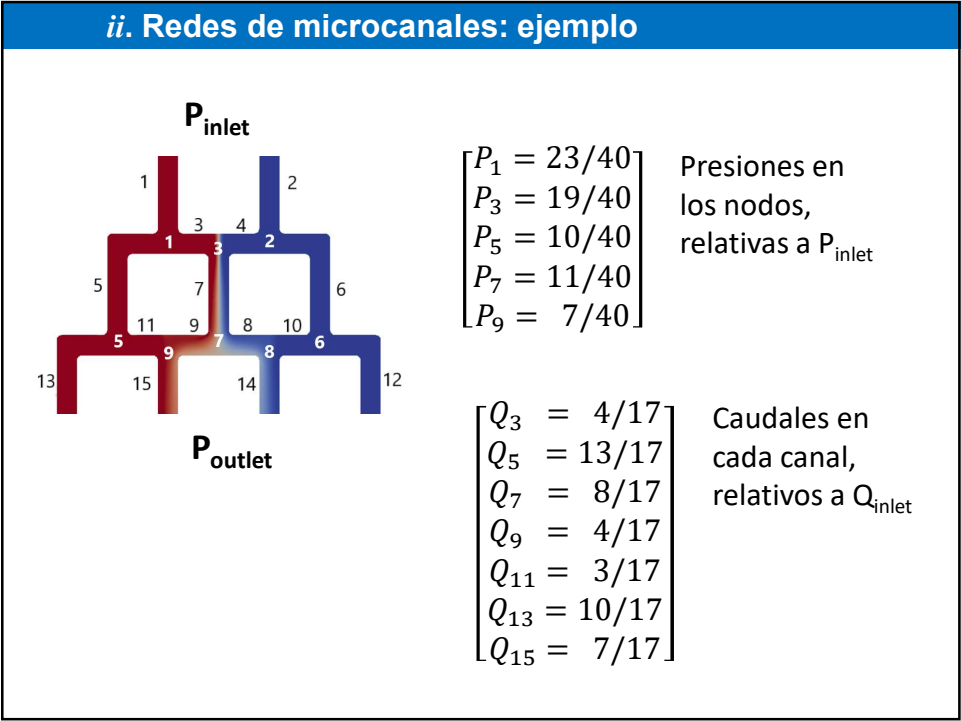
→ **M incógnitas**



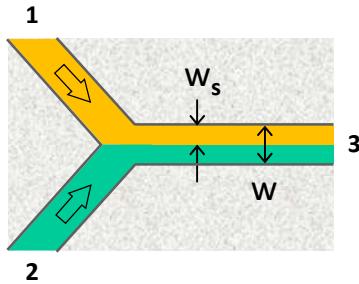
$$\sum_{i=1}^N Q_i = 0, \quad j = 1 \text{ a } M$$

$$\begin{bmatrix} \sum_{i=1}^N L_{11,i} & -L_{11,cc} \\ -L_{11,cc} & \sum_{i'=1'}^{N'} L_{11,i'} \end{bmatrix} \begin{bmatrix} P_c \\ P_{c'} \end{bmatrix} = \begin{bmatrix} \sum_{i \neq cc}^N L_{11,i} P_i \\ \sum_{i' \neq cc}^{N'} L_{11,i'} P_{i'} \end{bmatrix}$$





ii. Co-flujos, juntura Y



$$\mu_1 = \mu_2$$

$$\rho_1 = \rho_2$$

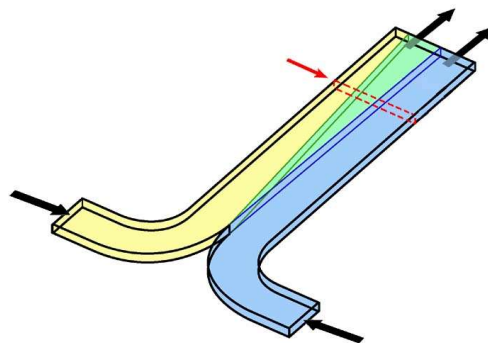
$$\frac{w_s}{w} = \frac{Q_1/Q_2}{1 + Q_1/Q_2}$$

$$\begin{aligned} P_3 &= P_{\text{atm}}, \\ L_1 &= L_2 = L_3 \\ 1/2 &\leq P_1/P_2 \leq 2 \end{aligned}$$



$$\frac{w_s}{w} = \frac{2 - P_2/P_1}{P_2/P_1 + 1}$$

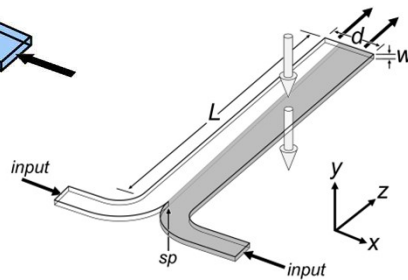
ii. Co-flujos, juntura Y: difusión



$$t_D \sim \Delta x^2/D$$

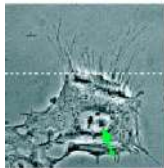
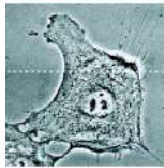
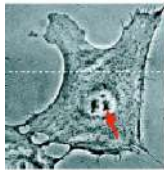
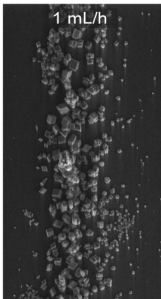
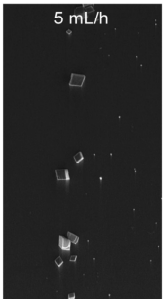
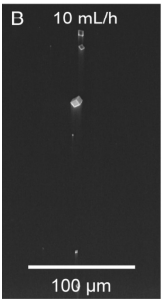
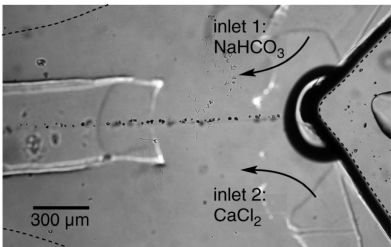
vs

$$t_r \sim z/u_z$$



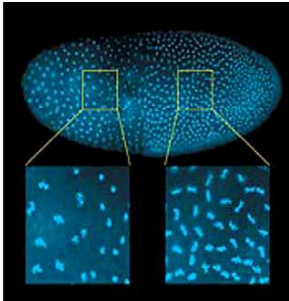
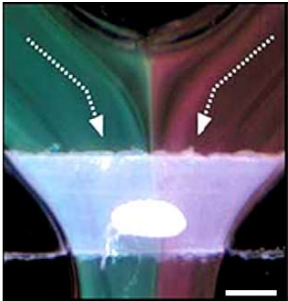
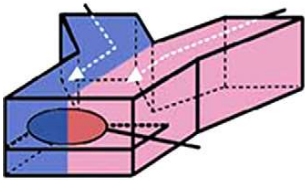
ii. Co-flujos, juntura Y: aplicaciones

Chia et al.,
J Chem Ed
88 (2011) 461



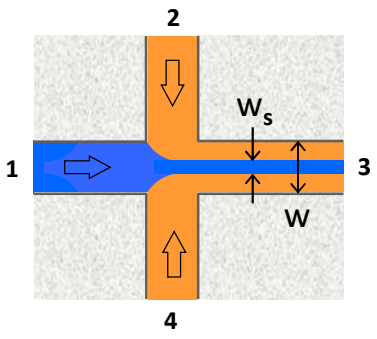
Takayama et al.,
Chem. Biol. **10** (2003) 123

ii. Co-flujos, juntura Y: aplicaciones



Lucchetta et al., *Nature* **434** (2005) 1134

ii. Co-flujos, juntura X

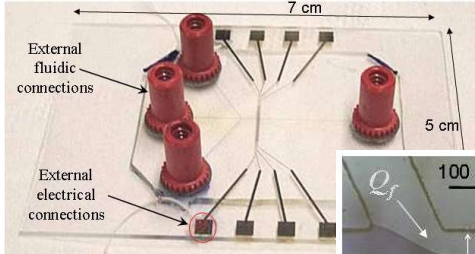


$$\mu_1 = \mu_2$$
$$\rho_1 = \rho_2$$
$$\frac{w_s}{w} = \frac{1}{1 + 2Q_{2,4}/Q_1}$$

$$P_3 = P_{atm},$$
$$L_1 = L_2 = L_3 = L_4$$
$$2/3 \leq P_{2,4}/P_1 \leq 3/2$$

$$\Rightarrow \frac{w_s}{w} = \frac{3/2 - P_{2,4}/P_1}{1/2 + P_{2,4}/P_1}$$

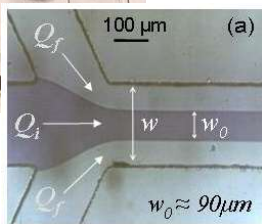
ii. Coflujos, juntura X: aplicaciones



External fluidic connections

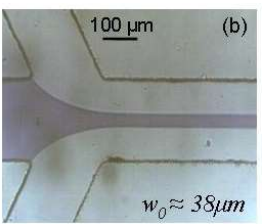
External electrical connections

Rodriguez et al.,
Microfluid Nanofluid **3** (2007) 171



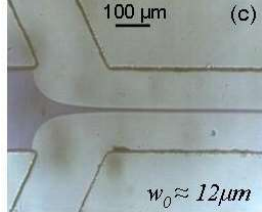
100 μ m (a)

$w_0 \approx 90 \mu$ m



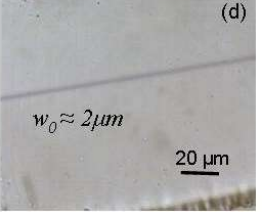
100 μ m (b)

$w_0 \approx 38 \mu$ m



100 μ m (c)

$w_0 \approx 12 \mu$ m

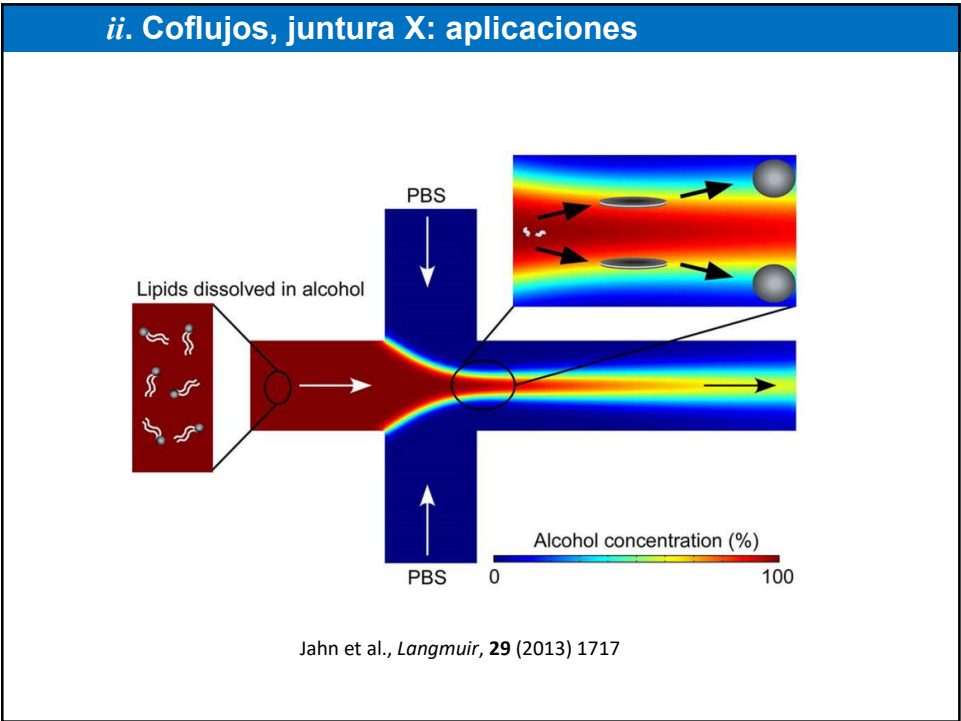
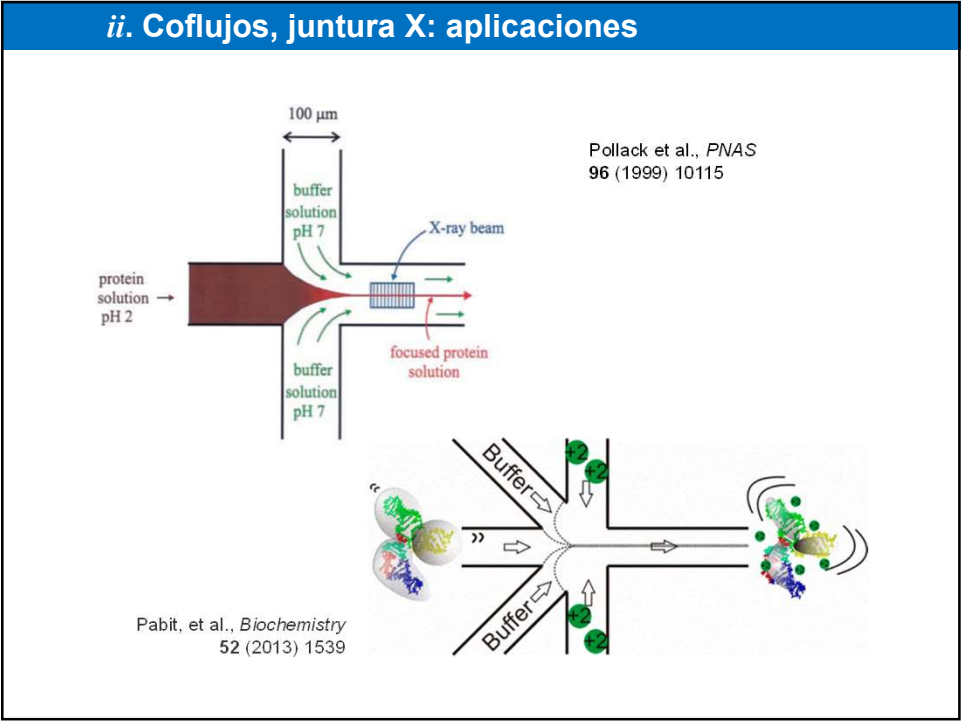


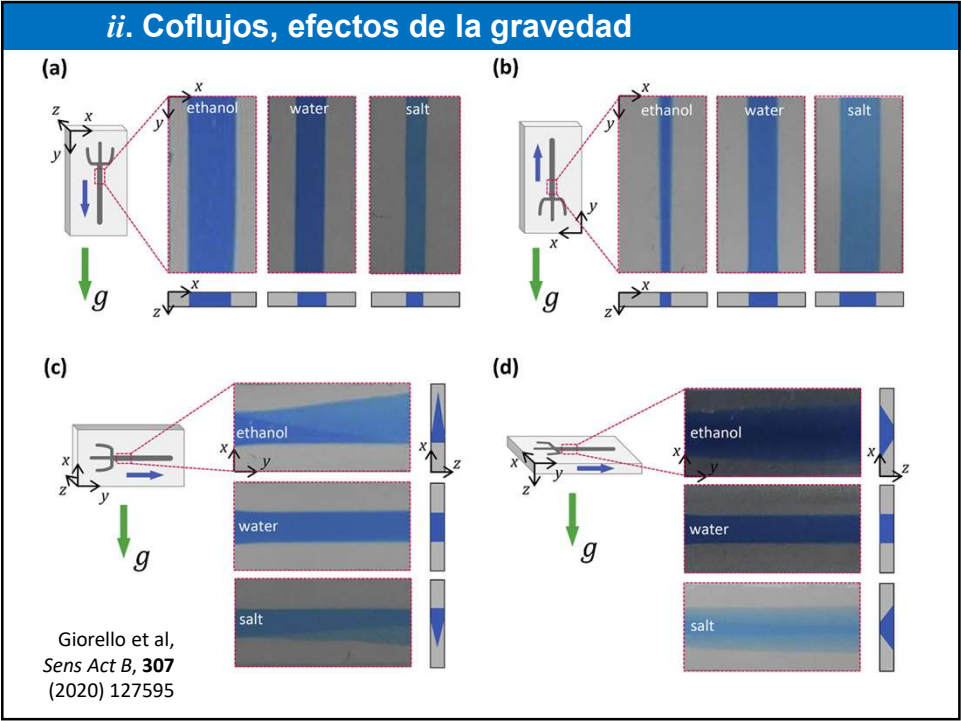
20 μ m (d)

$w_0 \approx 2 \mu$ m

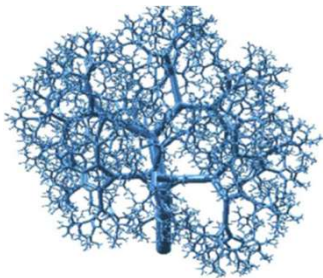
C. Berli

17



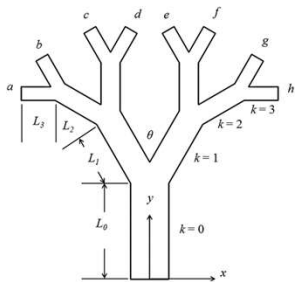


ii. Diseño para minimizar la resistencia



$$\frac{d_k}{d_{k+1}} = 2^{1/3}$$

1,26

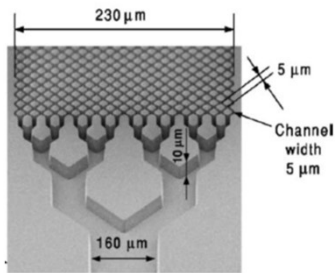


$$\frac{d_k^*}{d_{k+1}^*} = \frac{(1 + h/w)_k}{(1 + h/w)_{k+1}}$$

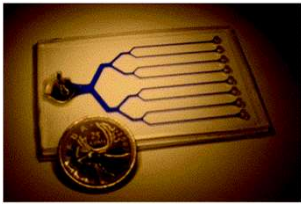
1,75

Emerson et al., *Lab Chip* **6** (2006) 447

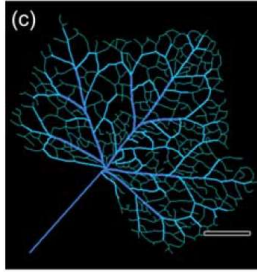
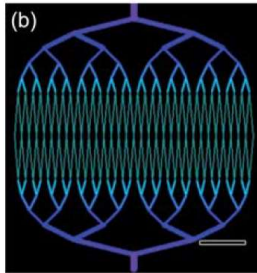
ii. Diseño para minimizar la resistencia



Slentz et al., *J. Chromatogr. A* **948** (2002) 225

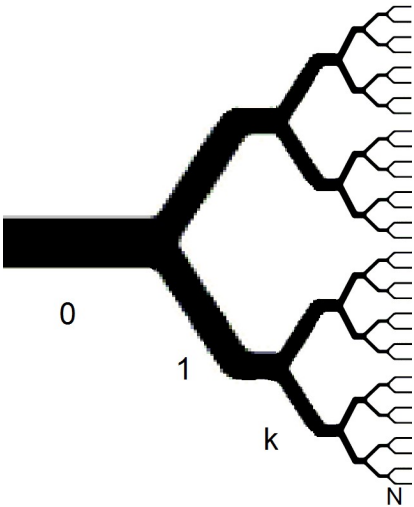


Young et al., *Lab Chip* **7** (2007) 1759



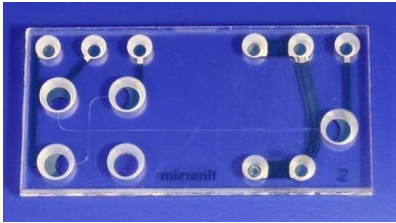
Wu et al., *Soft Matter* **6** (2010) 739

ii. Diseño para minimizar la resistencia



$$d_k = \frac{d_0}{2^{k/3}}$$
$$R_{\text{tot}} = \sum_0^N \frac{R_k}{2^k}$$
$$R_k = \frac{128\mu l_k}{\pi d_k^4}$$
$$R_{\text{tot}} = \frac{128\mu f_{ld}}{\pi d_0^3} (N + 1)$$

ii. Flujos 2D y 3D: simulación computacional (CFD)

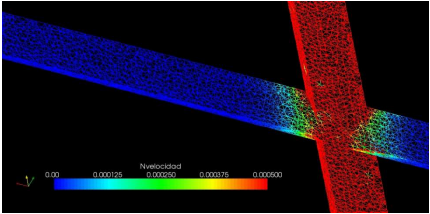
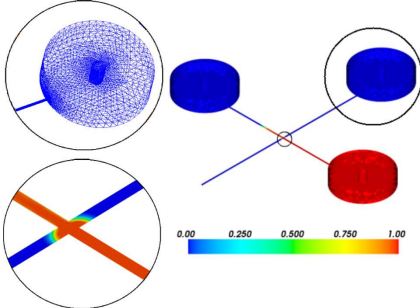


$\nabla \cdot \mathbf{u} = 0$

Newtoniano
incompresible
isotérmico

$$\rho \left(\frac{d\mathbf{u}}{dt} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla P + \mu \nabla^2 \mathbf{u}$$

Resolver: modelo geométrico
+ ecuaciones del modelo
+ parámetros del sistema



P. Kler, Tesis, 2011

